

EMPLOYMENT

- **Postdoctoral Researcher** (*incoming*) (June 2026 -)
Télécom Paris
- **Visiting Scientist** (February 2026 - May 2026)
R. C. Bose Centre for Cryptology and Security
Indian Statistical Institute, Kolkata.

EDUCATION

- **Indian Statistical Institute, Kolkata, India** (2019 - 2026)
Doctor of Philosophy (Ph.D) in Statistics.
 - Senior Research Fellow: July, 2021 - February, 2026
 - Junior Research Fellow: July, 2019 - July, 2021

Title: Statistical Guarantees of Deep Generative Models Involving Diverse Spaces: Generation Consistency and Robustness
Advisors: Prof. Swagatam Das, Prof. Probal Chaudhuri.
- **Indian Institute of Technology, Kanpur, India** (2017 - 2019)
Master of Science (M. Sc.) in Statistics.
- **St. Xavier's College, Kolkata, India** (2014 - 2017)
Bachelor of Science (B. Sc.) in Statistics.

RESEARCH INTERESTS

Theory of Deep Generative Models, High-dimensional Statistics, Optimal Transport and Alignment, Robust Statistics, Causal Inference, Differential Privacy.

PUBLICATIONS & PREPRINTS

Paper:  | Code:  | arXiv: 

- [1] **Anish Chakrabarty** and Swagatam Das. “Statistical Regeneration Guarantees of the Wasserstein Autoencoder with Latent Space Consistency”. In *Advances in Neural Information Processing Systems (NeurIPS)*, 34, pp.17098-17110, 2021. (**Spotlight**) 
- [2] **Anish Chakrabarty** and Swagatam Das. “On Strong Consistency of Kernel k -means: A Rademacher Complexity Approach”. In *Statistics & Probability Letters*, 182, p.109291, 2022. 
- [3] **Anish Chakrabarty** and Swagatam Das. “On Translation and Reconstruction Guarantees of the Cycle-Consistent Generative Adversarial Networks”. In *Advances in Neural Information Processing Systems (NeurIPS)*, 35, pp.23607-23620, 2022. 
- [4] Shounak Datta, Sankha Subhra Mullick, **Anish Chakrabarty**, and Swagatam Das. “Interval Bound Interpolation for Few-shot Learning with Few Tasks”. In *International Conference on Machine Learning (ICML)*, pp. 7141-7166. PMLR, 2023. 
- [5] Debanjan Dutta, **Anish Chakrabarty**, and Swagatam Das. “Lost in Translation: GANs’ Inability to Generate Simple Probability Distributions”. In *International Conference on Learning Representations (ICLR) Tiny Papers*, 2024.  *
- [6] Abhishek Kumar, **Anish Chakrabarty**, Sankha Subhra Mullick, and Swagatam Das. “Enhancing Contrastive Clustering with Negative Pair-guided Regularization”. In *Transactions on Machine Learning Research (TMLR)*, 2024. 

- [7] **Anish Chakrabarty**, Arkaprabha Basu, and Swagatam Das. “Information Preservation with Wasserstein Autoencoders: Generation Consistency and Adversarial Robustness”. In *Statistics and Computing*, vol 35, 114, 2025. 
- [8] **Anish Chakrabarty**, Sankha Subhra Mullick, and Swagatam Das. “Locally Robust Alignment Between Distinct Spaces”. In *Stat* 14, no. 3: e70093, 2025. 
- [9] Dhruv Sarkar, Aprameyo Chakrabartty, **Anish Chakrabarty**, and Swagatam Das. “On Relation-Aware Slicing in Cross-Domain Alignment”, In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026. *To appear.*  *
- [10] **Anish Chakrabarty**, Arkaprabha Basu, and Swagatam Das. “On Robust Cross Domain Alignment”, 2025. *Under review.*  
- [11] Abhishek Kumar, **Anish Chakrabarty**, and Swagatam Das. “DeepMALC: Integrating Mode Seeking with Deep Manifold Learning for Automatic Clustering”, 2025. *Revision submitted.* 
- [12] Debanjan Dutta, **Anish Chakrabarty**, Faizanuddin Ansari, and Swagatam Das. “On the Existence of Universal Simulators of Attention”, 2025. *Under review.* 
- [13] Shinjon Chakraborty, **Anish Chakrabarty**, and Swagatam Das. “Risk-Calibrated Treatment Effect Estimation”, 2026. *Under review.* *

ONGOING WORK

*(Projects mentored)

- [1] Dhruv Sarkar, Sankha Subhra Mullick, and **Anish Chakrabarty**. “Federated Estimation of Gromov-Wasserstein-type Distances”.
- [2] **Anish Chakrabarty**. “A Depth-based Estimator of Truncation Intensity in Robust Cross-Domain Alignment”.

AWARDS

- DST-INSPIRE Scholarship awarded by the Department of Science and Technology, Government of India. (2014 - 2019)
- Junior Research Fellowship (2019 - 2021), and Senior Research Fellowship (2022 - present) awarded by the Indian Statistical Institute.
- ISI Kolkata graduate student travel grant, 2024. (*to attend ICLR 2024 in Vienna, Austria*)
- Science and Engineering Research Board (SERB) travel grant awarded by the Department of Science and Technology, Government of India, 2024.

PROFESSIONAL SERVICES

- Organizing committee member of the Winter School on Deep Learning (2022 - 2026) hosted by the Electronics and Communication Sciences Unit, Indian Statistical Institute, Kolkata. ( <https://sites.google.com/view/wsd126>)
- Reviewer at IEEE Transactions on Neural Networks and Learning Systems (2023 onwards), Neural Networks (2023 onwards), IEEE Transactions on Cybernetics (2021 onwards), Pattern Recognition (2021 onwards), Applied Soft Computing (2022 onwards).
- Reviewer at NeurIPS, ICLR, ICML, CVPR, AAAI, ICCV, ECCV, AISTATS.

TECHNICAL SKILLS

R, Python, C/C++, Matlab, Minitab.

TALKS & PRESENTATIONS

- ISI-ISM-ISSAS Joint Conference 2022 (*Poster Presentation*): “On Strong Consistency of Kernel k-Means: A Rademacher Complexity Approach”.
- Course Lectures for WSDL: Statistical Learning Theory (2022), Introduction to Statistical Computing (2023), Generative Adversarial Networks (2023 - 2024), Probabilistic Diffusion Models (2024 - 2025), Variational Inference (2025), Cross-Domain Learning (2025).

CONFERENCES

- Lectures on Probability and Stochastic Processes (LPS) 2019, hosted by Theoretical Statistics and Mathematics Division, Indian Statistical Institute, Delhi.
- ISI-ISM-ISSAS Joint Conference 2022.
- Data Science: Probabilistic and Optimization Methods II, hosted by the International Centre for Theoretical Sciences (ICTS), TIFR, Bangalore.